EXPERIMENT 8 — LAB RECORD

Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

Bank Loan Application Form — User Permissions, Roles & Data Sharing Use Case

---

Submitted by

Mohid Ahmed F

Register Number: **241401057**

Aim

To configure user management and role-based access control (RBAC) within the Bank Loan Application Form application built in Zoho Creator, by creating permission sets, constructing a role hierarchy, and configuring a custom data-sharing rule that grants the loan officer role Read/Write access to loan application data owned by the applicant role.

Objectives

- To review the application's live dashboard, which summarises loan applications by loan type, loan tenure and employment type.
- To create two permission sets, Loan Administrator and Loan Applicant, alongside the default Read and Write sets under User Permissions.
- To construct the role hierarchy CEO → {loan officer, applicant, Guarantor, Auditor}.
- To configure a Data Sharing rule, Applicant_to_Loan_Officer, sharing loan application data from the applicant role to the loan officer role with Read/Write access.
- To understand how RBAC separates data ownership (who submits a record) from data visibility (who else may view or edit it).

Software / Tools Required

- Zoho Creator (web-based low-code application development platform)
- Bank Loan Application Form application
- Web browser with an active internet connection
- Zoho account with access to the application in Edit mode

Introduction / Theory

User Management in Zoho Creator

User management decides who can access an application and what each person may see and do within it. Users are added to an application and associated with permission sets (profiles) and roles; together these control access to forms, reports, pages and records. In the Settings area, the User Permissions section groups these controls under three tabs: Permissions, Roles and Data Sharing.

Role-Based Access Control (RBAC)

RBAC is an access-control model in which rights are assigned to roles instead of individual users, and users acquire rights through the roles they hold. It simplifies administration, supports the principle of least privilege, improves data security, and keeps the access policy consistent and easy to audit.

Key Concepts

Concept	Description
Users	Individuals added to the application who sign in to work with its forms, reports and data.
Permissions	Named permission sets (profiles) defining what a user may do with components. This application lists Loan Administrator, Loan Applicant, Read, and Write.
Roles	Positions in the organisational structure (CEO, loan officer, applicant, Guarantor, Auditor) that determine a user's place in the hierarchy and the visibility of records.
Role Hierarchy	A tree in which each role reports to another. A higher role can generally view the records of roles beneath it; the reverse does not apply.
Data Sharing	Share rules that extend record visibility beyond the hierarchy, from one role or user to another, with a defined level of access (Read Only or Read/Write).

Application Specifications

Parameter	Detail
Application Name	Bank Loan Application Form
Primary Forms	loan application form (loan applications)
Dashboard	Charts by loan type, loan tenure and employment type (Salaried, Self-Employed, Student), with Applications pending and Applications rejected summary tiles
Permission Sets	Loan Administrator, Loan Applicant, Read, Write
Role Hierarchy	CEO → {loan officer, applicant, Guarantor, Auditor}
Custom Sharing Rule	Applicant → loan officer, on the loan application form, Read/Write access

2241401057

Procedure

1. Sign in and locate the application

Signing in to Zoho Creator opens the home page listing the available applications; the application card, shown in Zoho Creator under the name Bank Loan Application Form, is located among them.

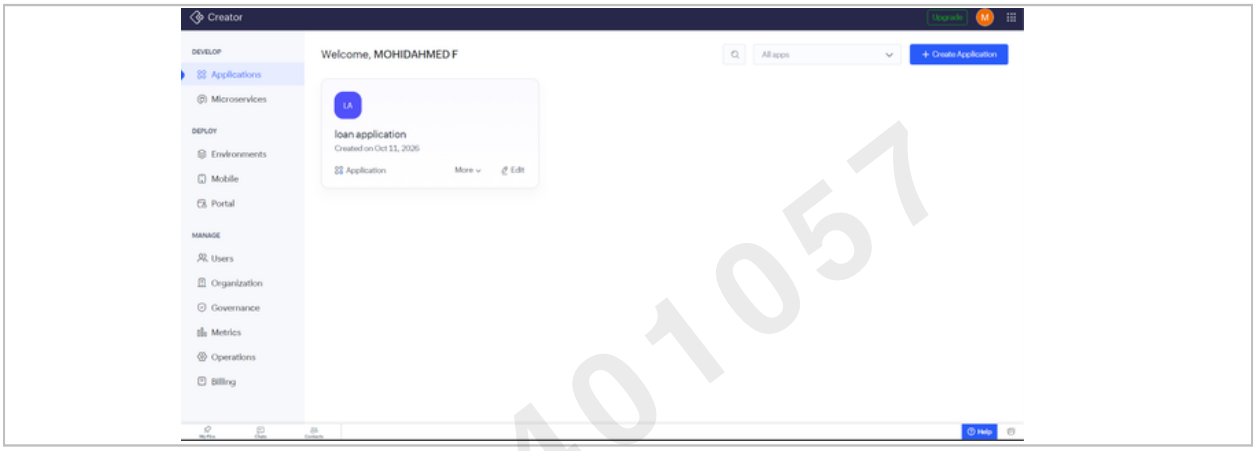


Fig. 8.1 — Zoho Creator Home — Application Listing

The home page lists the applications in the account, with the Bank Loan Application Form card alongside its Application and Edit shortcuts.

2. Open the dashboard

Opening the application loads its dashboard in Access mode, showing charts and summary tiles for the loan applications.

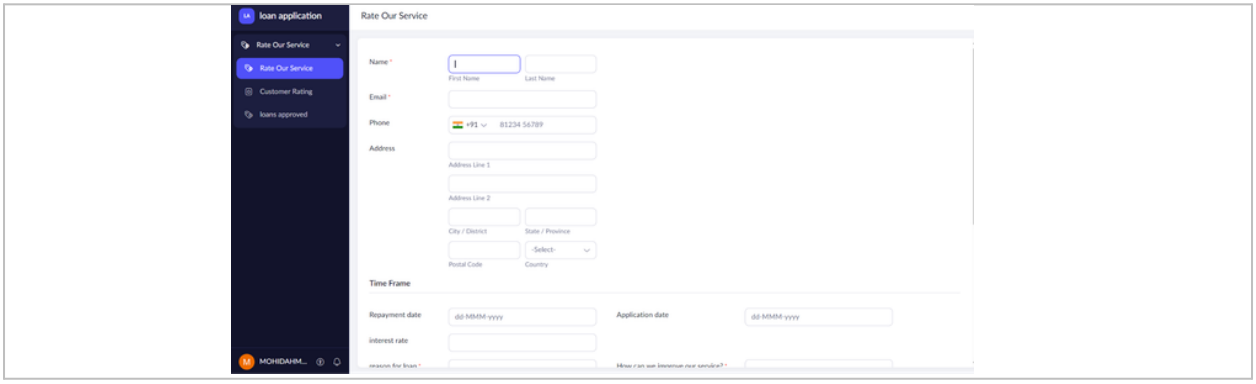


Fig. 8.2 — Application Dashboard (Access Mode)

The dashboard shows donut and pie charts and a bar chart summarising applications by loan type, loan tenure and employment type, along with the Applications pending and Applications rejected summary tiles.

2414401057

3. Enter Edit mode and open Settings

Selecting Edit this application switches to Edit mode, and the Settings tab is selected from the top navigation bar (Design, Workflow, Settings).

4. Open User Permissions

Under the PERMISSIONS group, User Permissions is selected, opening the Users page with the Permissions, Roles and Data Sharing tabs.

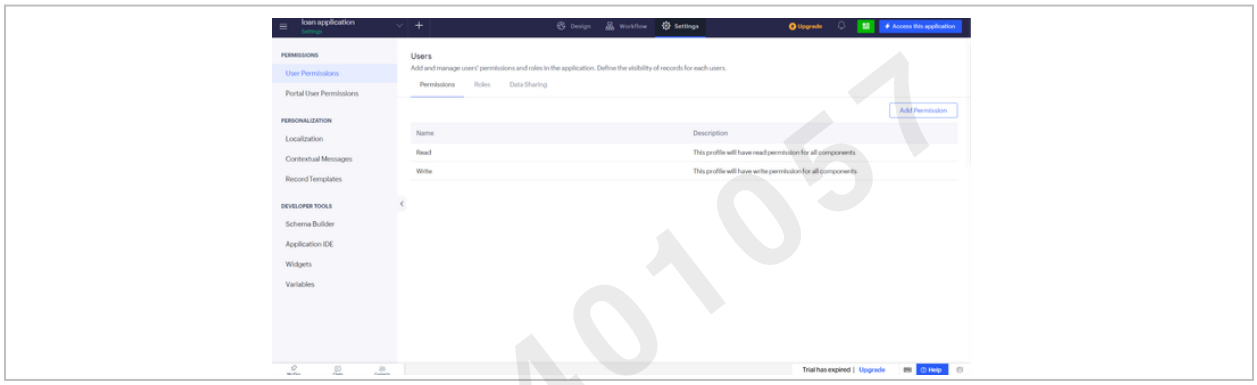


Fig. 8.3 — User Permissions — Permissions Section

The Permissions tab lists four permission sets: Loan Administrator (reviews loan applications, verifies submitted documents and approves or declines applications), Loan Applicant (submits loan applications and views their own application status), Read, and Write — each with a short description and an Add Permission option.

5. Build the role hierarchy

On the Roles tab, the role tree is arranged as CEO → loan officer, applicant, Guarantor, and Auditor, using Expand All, Collapse All, Manage Role Hierarchy and Add Role as needed.

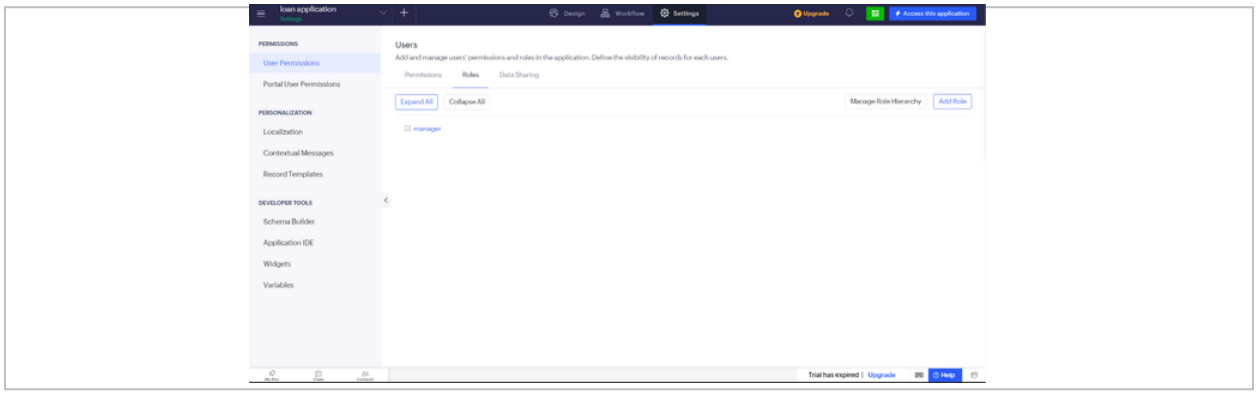


Fig. 8.4 — Roles and Role Hierarchy

The Roles tab shows the completed hierarchy: CEO at the top, with loan officer, applicant, Guarantor, and Auditor all reporting to the CEO as sibling roles.

6. Create a new share rule

On the Data Sharing tab, Add Rule is selected, the rule is named Applicant_to_Loan_Officer, the Form is set to loan application, and Records shared from is set to Roles → applicant.

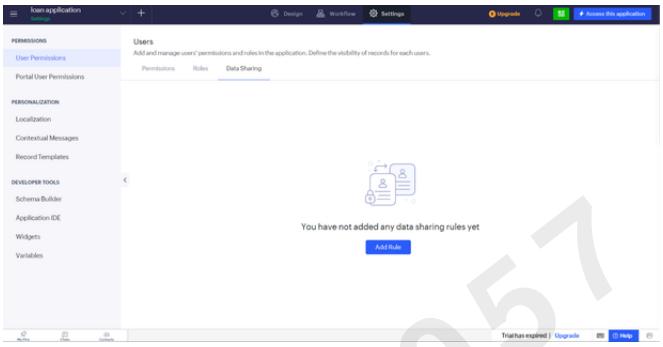


Fig. 8.5 — New Share Rule — Rule Configuration

The New Share Rule panel shows the rule Applicant_to_Loan_Officer on the loan application form, with Records shared from set to Roles → applicant, Records shared To set to Roles → loan officer, and Level of access set to Read/Write.

7. Save and verify the rule

Selecting Save adds the rule to the Data Sharing list, where the saved values can be checked.

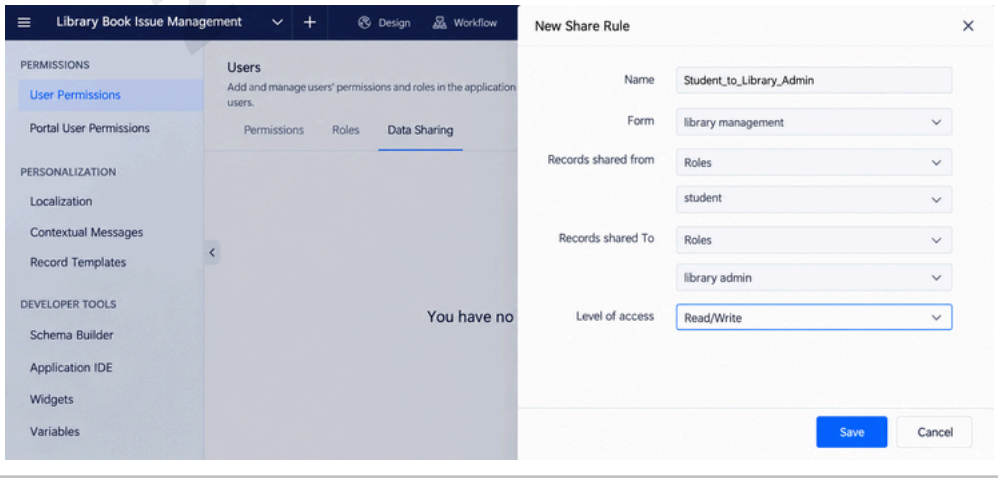


Fig. 8.6 — Configured Data Sharing Rule (Applicant → Loan Officer, Read/Write)

The Data Sharing list now shows the saved rule — Form: loan application, Shared From: R:applicant, Shared To: R:loan officer — with the Level of access set to Read/Write.

Data Sharing

The application follows a layered model: Form (loan application / loan requests) → Dashboard → Security Layer (Permissions, Roles, Data Sharing). The role hierarchy runs from the CEO (overall oversight) down to the loan officer, applicant, Guarantor, and Auditor roles, which sit side by side beneath it. By default the hierarchy gives only upward visibility: higher roles see the records of the roles below them, but not the reverse.

The custom data-sharing rule shares loan application data from the applicant role to the loan officer role at Read/Write level. Because the loan officer and applicant roles are siblings under the CEO, the loan officer cannot see applicant records through the hierarchy alone. The rule gives the loan officer an explicit, auditable Read/Write channel to review and update application details directly, keeps the scope directional (applicant → loan officer), documents exactly who can modify what in the Data Sharing list, and lets the loan officer act on records without needing elevated, system-wide Write permission.

RBAC Working

Level	Role	Intended Function
1 (Top)	CEO	Overall authority; able to view the records of every role reporting to it.
2	loan officer	Reviews loan applications, verifies submitted documents and approves or declines applications; gains Read/Write access to applicant-owned applications through the custom share rule.
2	applicant / Guarantor / Auditor	Applicants submit loan applications and view the status of their own applications; the Guarantor and Auditor roles work with records according to their assigned permission set.

2414O1057

- Users in the applicant, Guarantor, and Auditor roles work with records according to their assigned permission set.
- Because all four roles report directly to the CEO, higher-role visibility applies only to the CEO; the sibling roles do not automatically see each other's records.
- The Read/Write share rule grants the loan officer a controlled, explicit channel to applicant-owned loan applications.
- The permission sets (Loan Administrator, Loan Applicant, Read, Write) further limit the specific operations each user may carry out on each component.

Result

- The Bank Loan Application Form application, with its loan application form, was opened and its dashboard reviewed.
- Two custom permission sets, Loan Administrator and Loan Applicant, were added alongside the default Read and Write sets under User Permissions.
- The role hierarchy CEO → {loan officer, applicant, Guarantor, Auditor} was created.
- A data-sharing rule from applicant to loan officer with Read/Write access on the loan application form was configured and saved.

Conclusion

The experiment demonstrated how permissions, roles, role hierarchy and data sharing together form a role-based access control mechanism in Zoho Creator. Structured forms and a role-aware dashboard replace the paper forms and verbal requests of a manual bank counter with clean, consistent, and analysable data. The two-level hierarchy and the directional Read/Write share rule keep information channels controlled: the loan officer gains exactly the access needed to review and process applicant loan applications, while the CEO retains overall oversight and the applicant, Guarantor, and Auditor roles continue to own their respective records. The result is a secure, auditable and scalable access-control framework for the Bank Loan Application Form application.

241401057